$$\xrightarrow[\times(-1/2)]{Row2 = Row2} \begin{bmatrix} 1 & -3 & 2 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow[+Row3]{Row2 = Row2} \begin{bmatrix} 1 & -3 & 2 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow[-2\times Row2]{Row1 = Row1}$$

$$\begin{bmatrix} 1 & -3 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow[+3\times Row2]{Row1 = Row1} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Verification of whether the final matrix is in rref

1. Row 1, 2 and 3 are non-zero rows. Row 4 is a zero row. $\therefore$ The condition that all non-zero rows must be above the zero rows is True.
2. Row 1 Leading Entry = 1 in Column 1
   Row 2 Leading Entry = 1 in Column 2
   Row 3 Leading Entry = 1 in Column 3
   $\therefore$ The condition that for each row, the leading entry must be in a column to the right of the columns of the leading entries of all rows above it is True.
3. Row 2 Column 1 = Row 3 Column 1 = Row 4 Column 1 = 0
   Row 3 Column 2 = Row 4 Column 2 = 0
   Row 4 Column 3 = 0
   $\therefore$ The condition that for each column having a leading entry, all elements below the leading entry position must contain 0 is True.
4. The condition that each leading entry must be 1 is True.
5. Row 1 Column 3 = Row 2 Column 3 = 0
   Row 1 Column 2 = 0
   $\therefore$ The condition that for each column having a leading entry, all elements above the leading entry position must contain 0 is True.

$\therefore$ The matrix $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ is in rref. $\square$

Each column of A has a leading entry. $\therefore$ By defn, all columns of A are pivot columns. By defn, there are no free variables. By Theorem 2, the system of equations has a unique (trivial) soln. $\therefore$ The columns of matrix A form a linearly independent set. $\square$